STADIO HIGHER EDUCATION

Data Science in Finance (Elective)

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SS3 — Part B (Report)

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1. Introduction

Lending is central to financial institutions, and the risk that borrowers fail to repay is a persistent threat to their stability. Araya et al. note that traditional credit assessment depends heavily on the experience of the financial analyst and on information supplied largely by the debtor. Machine learning has been proposed to improve this process, and the case study under review applies several such models to ten years of loan records from a private commercial bank in Ethiopia.

The study's stated aim is to predict whether a loan will be repaid, comparing ensemble methods with a traditional logistic regression model. Its two research questions concern how the data should be prepared and which algorithm performs best for credit risk prediction. The analysis below responds to seven questions, and each question is answered under its own heading. The report concludes by combining these sections together and assessing the case study.

2. Issues that could compromise reproducibility

Several features of the case study limit its reproducibility. The authors state that no datasets were generated or analysed for the study, and the data is an extract from one private bank's core banking system. Therefore, the records cannot be obtained, and the analysis cannot be repeated. The reported dataset description is also internally inconsistent. The abstract gives over 47,000 loans, the data collection section gives 47,645 records, and the preprocessing section gives more than 47,600. The number of features is likewise stated as 16 and, later, as over 17.
A more substantive concern concerns the order of operations. Figure 2 places class balancing before the train and test split, whereas the text describes the training set as balanced. If resampling preceded the split, synthetic minority-class records could enter the test data, which would inflate the reported performance. The hyperparameters are also absent: the best estimators in Table 2 are not visible in the document, and CatBoost is described as reporting its best iteration rather than its parameters. Customer Id appears among the feature importance results but not in the variable list of Table 1, so the exact model inputs cannot be confirmed.

3. What decision or process is being improved, and for whom

The process being improved is the lending decision: whether a loan applicant is likely to repay. The authors describe the 5C principle as a conventional approach whose reliance on individual judgement carries major limitations. The study frames the problem in financial terms. Using the Development Bank of Ethiopia as an illustration, it reports non-performing loans of 17 billion Birr, or 36% of the portfolio. The principal beneficiaries are therefore the lending institutions, which the authors expect to reduce financial risk while increasing revenue. Applicants are affected indirectly, since the scoring models are presented as a means of making credit decisions faster and more consistent.

4. Data records, missing values and sensitivity

The dataset comprises 47,645 loan records from fiscal years 2012 to 2021. Each record describes one loan and its borrower, and carries a payment status classified as default or non-default. Before resampling, 558 loans are defaults and 42,323 are non-defaults, so defaults make up about 1.3% of the data. The variables fall into five groups: general, personal, loan, collateral and target, with location also recorded. Age and Collateral Type contain missing values, which are imputed with the mean and mode respectively. Gender, age, age range, country and Customer Id are sensitive, and the authors state that debtors' names were not disclosed.

Table 1 (report): Data records and variables

Group	Variable	Type / values	Description	Sensitivity
Time & Period	Fiscal year, Disbursement date, Maturity date	Number, 2012–2021	Loan origination and repayment timeline information	Low
Borrower Demographics	Gender, Age Age range	Categorical, Numeric	Personal characteristics of the borrower	High (personal)
Loan Characteristics	Sector code, Sector description, Disbursed amount, Loan term (months), Loan term (ST/MT/LT), Outstanding principal	Numeric, Categorical	Details relating to the type, size, duration, and outstanding balance of the loan	Medium (Financial)
Collateral Information	Collateral type, Collateral lien amount	Categorical, Numeric	Assets pledged as security and their associated value	Medium (Financial)
Credit Risk Outcome (Target)	Credit risk status, Credit risk code	Categorical, Binary	Loan repayment performance used as the prediction target	Medium
Geographic Information	Country	Categorical	Country associated with the borrower	High (Location)
Identifiers	Customer ID	Identifier	Unique borrower identifier used for linking records and feature analysis	High (Identifier)

Source: case study Table 1, with additions from the article text. The sensitivity ratings are this report's own judgement.

Two inconsistencies affect how the data can be read. Table 1 states that credit risk is grouped so that loss is coded 1 and all other statuses are coded 0. The text, however, codes bad risk as 0 and good risk as 1. In addition, Customer Id is absent from Table 1 yet appears in the feature importance results, so its status as a model input and its sensitivity are unresolved.

5. Modelling method and why it suits the problem

The principal method is the CatBoost Classifier, a gradient-boosted tree model applied as a supervised classifier that separates good from bad loans. The authors also test logistic regression, decision tree, random forest, histogram gradient boosting, XGBoost and LightGBM. The choice of tree-based

ensembles is justified by the nature of the data. The financial records are tabular, mix categorical and numerical fields, contain missing values and display complex relationships. The authors argue that such ensembles handle these conditions well and resist overfitting.
CatBoost is singled out because it processes categorical variables directly, which reduces the need for transformation, and because its boosting procedure concentrates on cases that are hard to predict. The authors attribute its superior performance to these properties.

6. Performance evaluation, baseline and measures

The baseline is logistic regression, the traditional model against which the ensemble methods are compared. The measures are weighted precision, recall, F1-score and accuracy derived from confusion matrices, together with ROC-AUC under 10-fold cross-validation. Differences in AUC are tested using DeLong's test, and differences between confusion matrices using McNemar's test. Table 6 summarises the results.

Model	ROC-AUC (CV = 10)	Weighted F1	Weighted accuracy
CatBoost Classifier	87	0.93	0.90
XGBoost Classifier	86	0.92	0.88
Random Forest Classifier	84	0.91	0.85
LGBM Classifier	83	0.89	0.82
Decision Tree Classifier	82	0.91	0.86
Hist Gradient Boosting Classifier	76	0.80	0.68
Logistic Regression (baseline)	57	0.60	0.44

Logistic regression has the lowest AUC, F1-score and accuracy of all models. The evaluation nonetheless contains a notable limitation. The test set holds 4,764 loans, of which 62 are defaulters and 4,702 are non-defaulters. Labelling every loan as good would therefore yield accuracy of about 98.7% (4,702 ÷ 4,764), which exceeds the 90% reported for CatBoost. The authors themselves acknowledge that accuracy is unreliable when the data are unbalanced. This comparison is the author of this report's calculation rather than a figure in the paper. Similarly, CatBoost's precision for the defaulter class is 0.10, implying that about nine in ten loans flagged as defaults were in fact repaid. This too is a reading of the reported figures.

7. How the model's output becomes an action

The model produces a probability for each loan request. For one example, CatBoost assigns a 99% probability of a good loan and a 1% probability of a bad loan. The article does not, however, set a decision threshold, describe an approval workflow or name the party responsible for acting on the output. The authors recommend CatBoost for predicting creditworthiness, but the translation of a prediction into a credit decision is left undefined.

A plausible pathway, offered here as a suggestion rather than as the authors' position, would route high-probability good loans to normal approval, decline or refer high-probability bad loans to a credit officer, and send borderline cases for manual review. The LIME explanations, which identify drivers such as collateral type and loan term, could support that review. Choosing the threshold would require weighing

the cost of wrongly rejecting a creditworthy borrower against the cost of approving a defaulter, a trade-off the paper does not examine.

8. Who is affected by failure, and how it would be detected

If the model fails, two groups bear the consequences. Creditworthy applicants are affected when they are wrongly classified as defaulters. The error analysis finds that these false positives were fewer and occurred mainly where collateral information was missing or inconsistent. Given the defaulter precision of 0.10, a large share of the loans flagged as bad would in fact have been repaid. The bank is affected when defaulters are approved. The authors find that most misclassifications are of this kind, typically involving borderline credit scores, moderate disbursed amounts or shorter tenures. A defaulter recall of 0.84 implies that roughly ten of the 62 defaulters in the test set were missed. The authors place these losses within a wider concern about capital erosion and financial instability.

Detection would be delayed, because repayment status is the ground truth. A loan moves to Special Mention after 30 days overdue and to Loss after 180 days, so failures become visible only after a lag. The authors detect model weaknesses through confusion matrices, error analysis and LIME explanations. In operational use, predictions should be compared with later repayment outcomes and the input data monitored for drift. The data span 2012 to 2021, and the paper reports no test for drift. This recommendation is the author of this report's own.

9. Conclusion

This report has examined the case study across seven questions set in the Part B brief. The study addresses a real problem: loan defaults threaten the stability of Ethiopian private banks, and the authors propose machine learning as an alternative to judgement-based assessment. Their headline finding is that CatBoost achieves an AUC of 0.87, an F1-score of 0.93 and an accuracy of 0.90, outperforming logistic regression, whose AUC is 0.57.

Taken together, the answers point to a clear distinction between the study's apparent strength and its verifiability. The modelling choices are justified by the characteristics of tabular financial data, and the evaluation uses a sensible range of measures with statistical comparison of models. However, the study cannot presently be reproduced. The dataset is unavailable, the record and feature counts disagree across sections, the ordering of resampling and splitting is unclear, and the tuned hyperparameters are not reported. The evaluation also overstates performance in a way the authors do not acknowledge. Accuracy is high largely because most loans are good, and the defaulter precision of 0.10 shows that many flagged loans would have been repaid.

The study is also incomplete as a decision tool. It supplies a probability, but not a threshold, an approval process or an account of who is harmed when the model errs. Its own error analysis shows that the most costly mistakes, defaulters approved as good, occur among borderline cases, and it recommends behavioural data to reduce such errors. A reasonable next step would be to address the reproducibility gaps, report the chosen threshold and its costs, and monitor the model against realised repayment outcomes.

Reference list

Araya, F.T., Assefa, B.G., Mohapatra, S.K. and Nanda, C.B. (2025) 'Sustainable credit risk prediction: applying machine learning for responsible lending', *Journal of Transformative Technologies and Sustainable Development*, 9, article 12. Available at: https://doi.org/10.1007/s41314-025-00080-7 (Accessed: 28 October 2026).